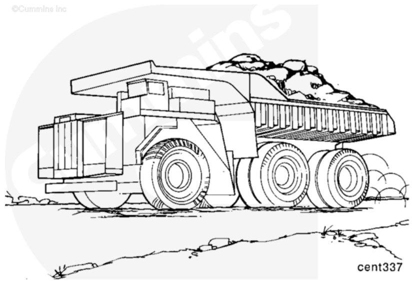
Trainings

General Information

The RS422 datalink circuit is used by vehicle systems to communicate with the CENSE™ ECM.

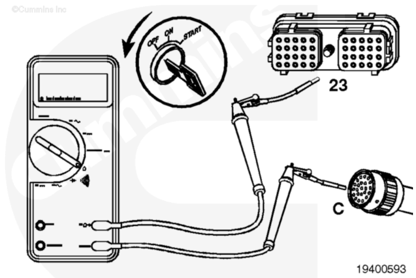


Resistance Check

Remove the CENSE™ harness ECM A connector from the ECM. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/81/81-019-043.html). Disconnect the OEM harness from the CENSE™ 23-pin OEM connector.

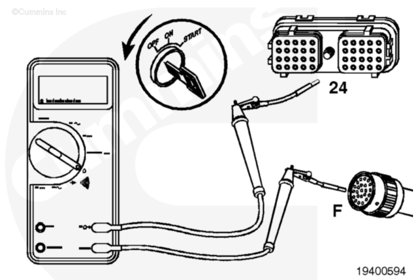
Use test leads, Part No. 3822758, on the ECM connector and Part No. 3824811 on the 23-pin Deutsch connector. Turn the keyswitch OFF.

Measure the resistance from pin 23 of the engine harness connector to pin C of the 23-pin Deutsch connector. The multimeter **must** show a closed circuit (10 ohms or less). If the circuit is **not** closed, repair or replace the CENSE™ harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).

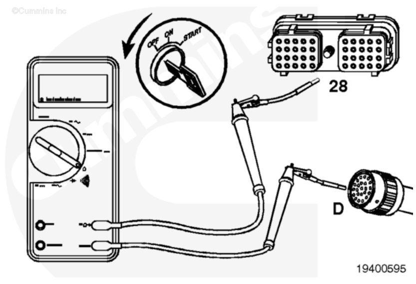


Measure the resistance from pin 24 of the CENSE™ harness connector to pin F of the 23-pin Deutsch connector. The multimeter **must** show a closed circuit (10 ohms or less).

If the circuit is **not** closed, repair or replace the CENSE™ harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).

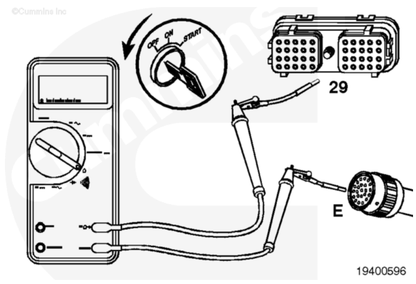


Measure the resistance from pin 28 of the CENSE™ harness connector to pin D of the 23-pin Deutsch connector. The multimeter **must** show a closed circuit (10 ohms or less).



If the circuit is **not** closed, repair or replace the CENSE™ harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).

Measure the resistance from pin 29 of the CENSE™ harness connector to pin E of the 23-pin Deutsch connector. The multimeter **must** show a closed circuit (10 ohms or less).

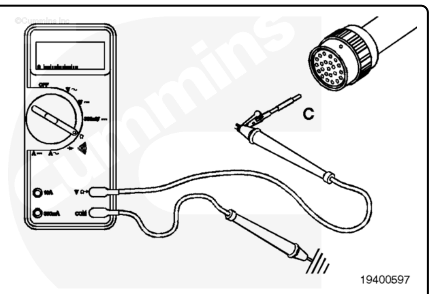


If the circuit is **not** closed, repair or replace the CENSE™ harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).

If all measurements are within specifications, the OEM harness **must** be checked. Refer to the OEM manual.

Check for Short Circuit to Ground

Turn the keyswitch OFF. Disconnect the 23-pin Deutsch OEM connector. Disconnect the ECM A and B connectors.

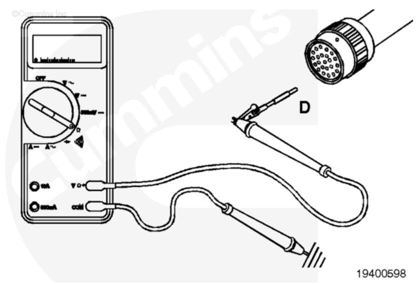


Use test lead Part No. 3824811 for the 23-pin Deutsch connector.

Measure the resistance from pin C of the 23-pin Deutsch connector to the engine block ground. The multimeter **must** show an open circuit (100k ohms or more).

If the circuit is **not** open, repair or replace the CENSE™ harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).

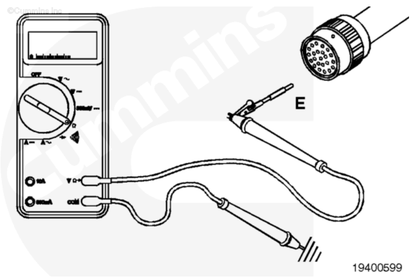
Measure the resistance from pin D of the 23-pin Deutsch connector to the engine block ground. The multimeter **must** show an open circuit (100k ohms or more).



If the circuit is **not** open, repair or replace the CENSE™ harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).

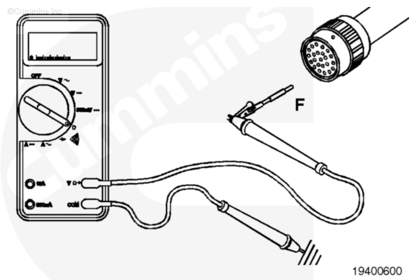
Measure the resistance from pin E of the 23-pin Deutsch connector to the engine block ground. The multimeter **must** show an open circuit (100k ohms or more).

If the circuit is **not** open, repair or replace the CENSE™ harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).



Measure the resistance from pin F of the 23-pin Deutsch connector to the engine block ground. The multimeter **must** show an open circuit (100k ohms or more).

If the circuit is **not** open, repair or replace the CENSE™ harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).



Check for Short Circuit from Pin to Pin

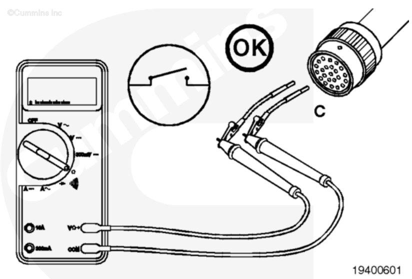
Deutsch Connector

Turn the keyswitch OFF. Disconnect the 23-pin Deutsch connector from the OEM harness.

Use test lead, Part No. 3824811, for the 23-pin Deutsch connector.

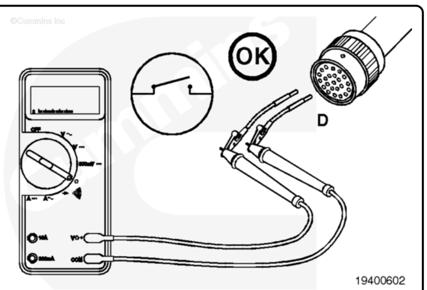
Measure the resistance from pin C to all other pins in the connector. The multimeter **must** show an open circuit (100k ohms or more).

If the circuit is **not** open, repair or replace the CENSE™ harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).

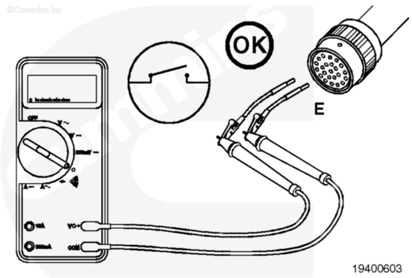


Measure the resistance from pin D to all other pins in the connector. The multimeter **must** show an open circuit (100k ohms or more).

If the circuit is **not** open, repair or replace the CENSE™ harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).

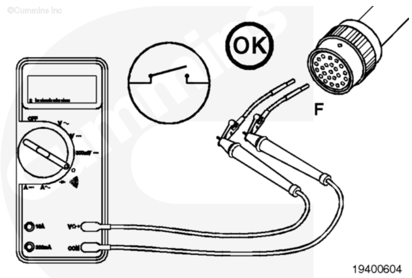


Measure the resistance from pin E to all other pins in the connector. The multimeter **must** show an open circuit (100k ohms or more).



If the circuit is **not** open, repair or replace the CENSE™ harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).

Measure the resistance from pin F to all other pins in the connector. The multimeter **must** show an open circuit (100k ohms or more).



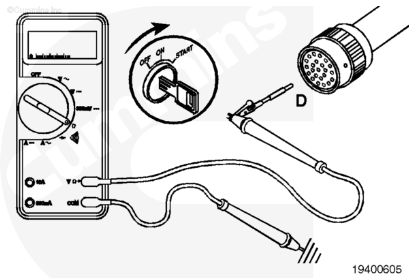
If the circuit is **not** open, repair or replace the CENSE™ harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/81/81-019-043.html).

If all measurements are within specifications, the OEM harness **must** be checked for a short circuit from pin to pin. Refer to the OEM manual.

Voltage Check

⚠ CAUTION ⚠

To avoid pin and harness damage, use test lead, Part No. 3824811.



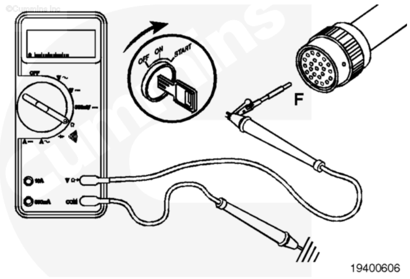
Disconnect the 23-pin Deutsch connector from the OEM harness.

Select the VDC function on the multimeter. Turn the keyswitch ON.

Measure the voltage from pin D of the 23-pin Deutsch connector to the engine block ground. The multimeter **must** show 0 to 3 VDC.

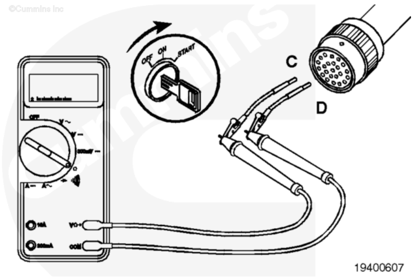
Measure the voltage from pin F of the 23-pin Deutsch connector to the engine block ground.

The multimeter **must** show 0 to 3 VDC.



Measure the voltage from pin C to pin D on the 23-pin Deutsch connector.

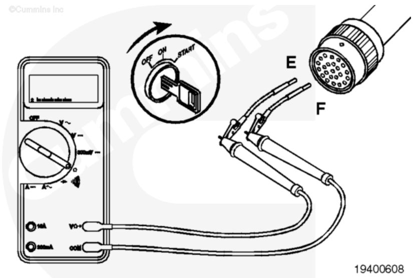
The multimeter **must** show 0 to 3 VDC.



Measure the voltage from pin E to pin F on the 23-pin Deutsch connector.

The multimeter **must** show 2 to 8 VDC.

If all measurements are within specifications, the OEM harness **must** be checked. Refer to the OEM manual.



Last Modified: 26-Aug-2003